

TWF2 Knockdown Reduces SNCA Expression with High Specificity

A Computational Perturbation Screen for Indirect Alpha-Synuclein Regulators

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1 Background

Parkinson's disease is the second most common neurodegenerative disorder, marked by the loss of dopaminergic neurons and the accumulation of alpha-synuclein into Lewy bodies.

Alpha-synuclein dosage is genetically linked to disease: duplication or triplication of the SNCA locus causes dominant Parkinson's, and the A30P, E46K, and A53T mutations cause early-onset familial disease.

Lowering SNCA is therefore a therapeutic aim, but selective knockdown is difficult. Antisense and RNAi must cross the blood-brain barrier and can trigger compensatory changes; here, direct SNCA knockdown altered 39 additional genes.

We therefore screened for an indirect regulator: a gene whose knockdown lowers SNCA downstream while perturbing few other transcripts. Actin regulators are plausible candidates, since alpha-synuclein binds actin and actin-severing proteins protect neurons in animal models.

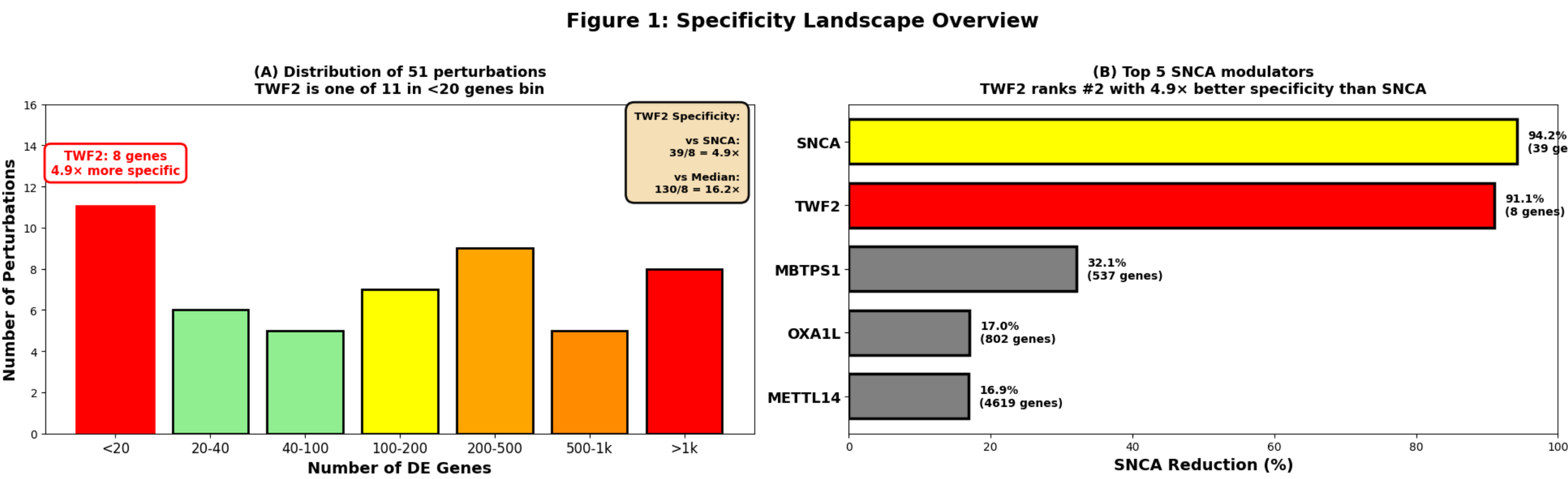


Figure 1. Specificity landscape. (A) The 51 perturbations binned by the number of genes each significantly changed; TWF2 falls among the 11 that affect fewer than 20 genes. (B) The five perturbations reducing SNCA most. TWF2 ranks second at 91.1%, below direct SNCA knockdown, at 4.9x greater specificity.

HYPOTHESIS

A specificity-first screen, which ranks perturbations by the number of genes they affect and then tests the most specific ones for SNCA reduction, should identify indirect regulators with a better safety profile than direct SNCA knockdown.

2 Methods

Dataset. Arc Institute Virtual Cell Challenge: ~300,000 H1 stem cells with CRISPRi knockdown plus 38,176 controls. We analyzed 51 perturbations (221,273 cells).

Processing. scanpy v1.9.6; normalized to 10,000 counts per cell, then log1p. Genes in under 5% of controls dropped, leaving 11,595.

Differential expression. Wilcoxon rank-sum test vs. controls with Benjamini-Hochberg correction. Changed = adjusted $p < 0.1$ and $|\log_2 \text{fold-change}| > 0.4$.

Specificity. Perturbations ranked by genes changed; fewer means more specific.

Safety. gnomAD pLI: ≥ 0.9 cannot tolerate loss of function, < 0.1 can.

3 Results

91.1%

SNCA mRNA reduction

8

total genes affected

4.9x

more specific than SNCA

Perturbation specificity spans three orders of magnitude. Across 51 perturbations, the number of significantly changed genes ranged from 1 to 4,028 (median 43). Eleven of 51 changed fewer than 20 genes and were carried forward for SNCA screening.

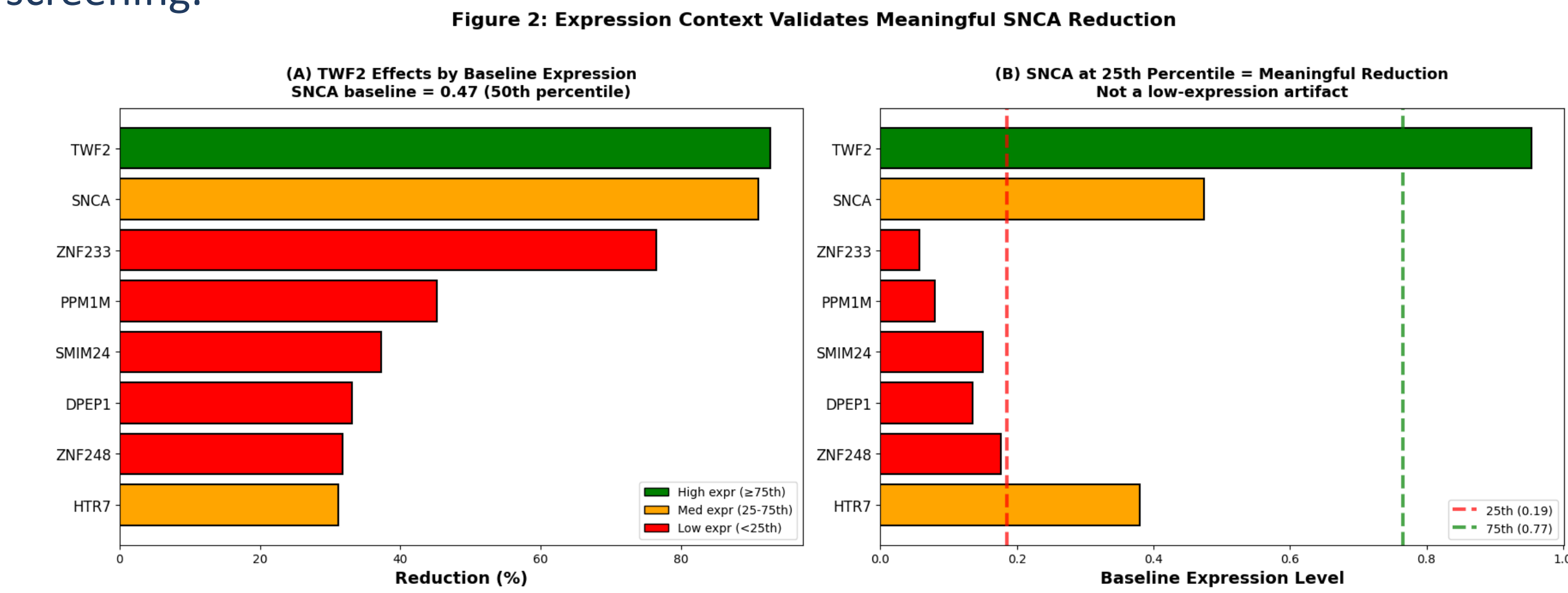


Figure 2. Baseline expression context for the SNCA reduction. (A) Percent reduction for the eight TWF2-affected genes, colored by baseline expression tier. (B) Baseline expression level for the same genes, showing that SNCA is expressed at a moderate level rather than near the detection floor.

TWF2 knockdown reduces SNCA by 91% and affects 8 genes. TWF2 knockdown reduced SNCA mRNA by 91.1% ($\log_2 \text{FC} = -3.54$, adj. $p < 10^{-100}$), against 94.2% for direct SNCA knockdown, while altering 8 genes versus 39. SNCA sits at the 50th expression percentile, so the 91% drop reflects a large absolute loss of transcript.

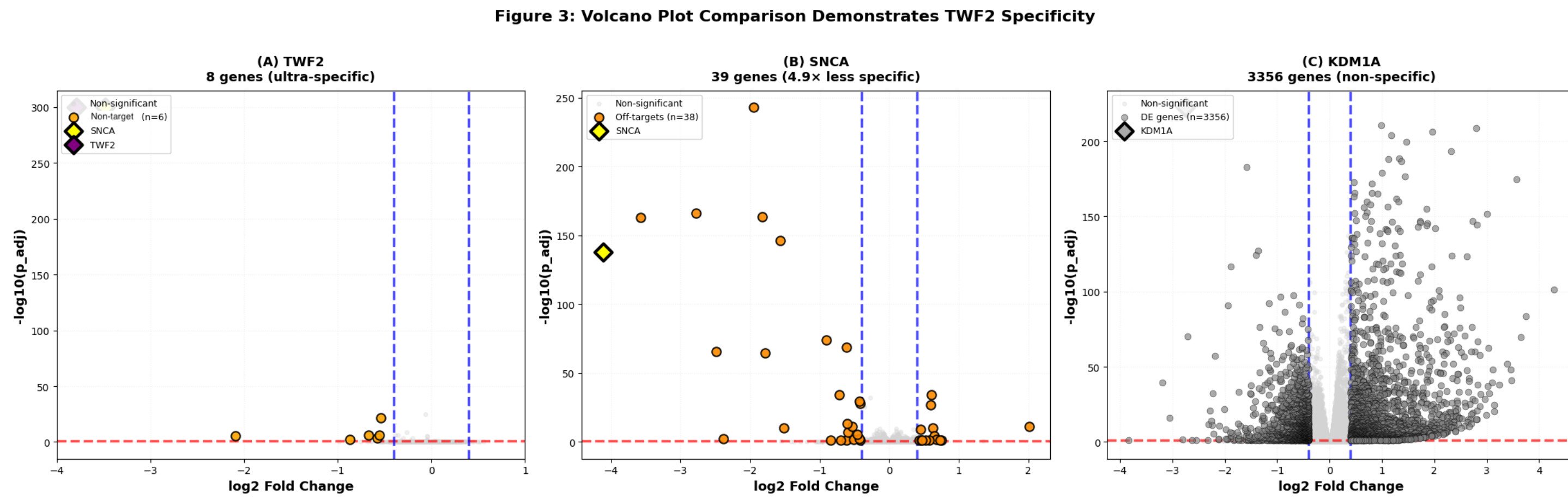


Figure 3. Volcano plots for three perturbations. Each point is a gene; points beyond the dashed thresholds (adj. $p < 0.1$, $|\log_2 \text{FC}| > 0.4$) are significant. (A) TWF2 knockdown alters 8 genes. (B) Direct SNCA knockdown alters 39. (C) A non-specific perturbation alters several thousand.

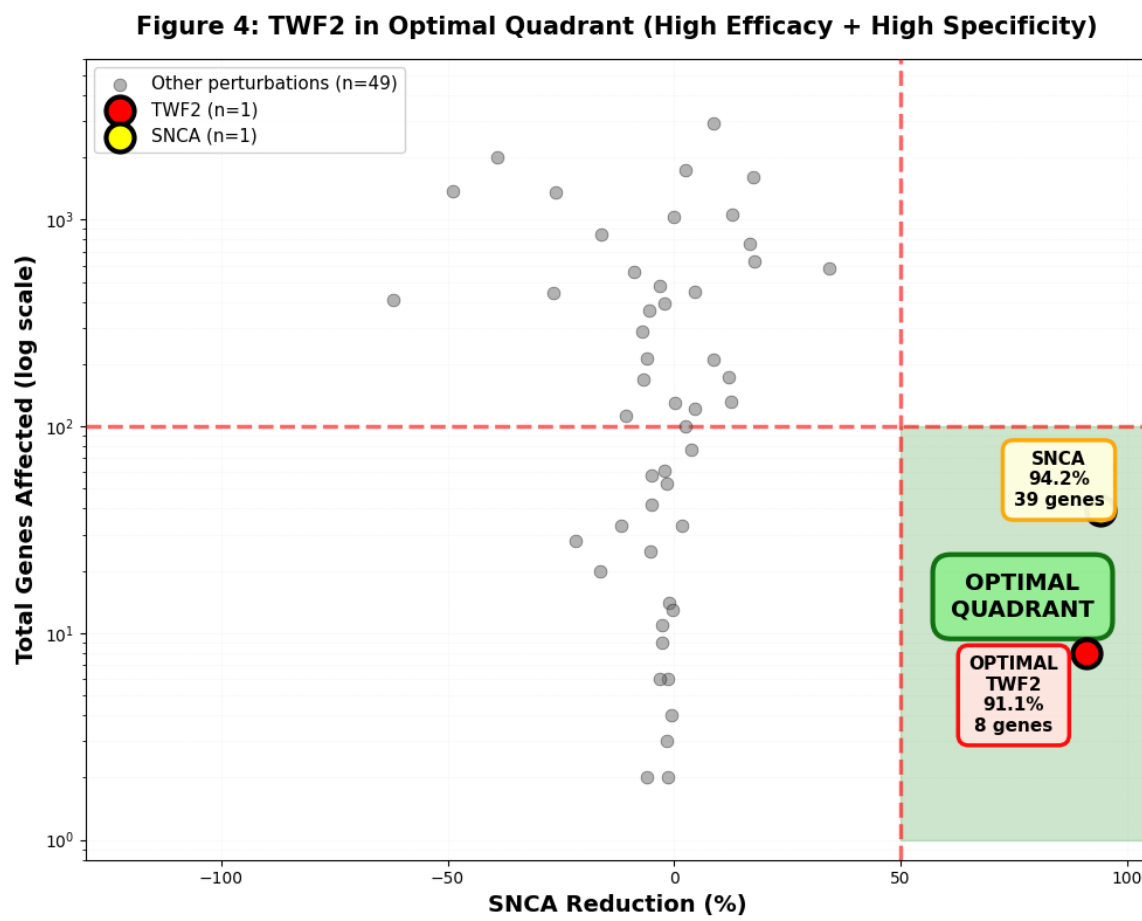


Figure 4. Efficacy versus specificity across all 51 perturbations. SNCA reduction is plotted against total genes affected (y-axis, log scale). TWF2 (red) is the only perturbation exceeding 90% reduction while affecting fewer than 10 genes.

4 Specificity & Safety

TWF2 and SNCA knockdowns act through distinct gene sets. Beyond the targets, TWF2 changed 6 genes and SNCA changed 37, with HTR7 (serotonin 5-HT7 receptor) the only shared gene. TWF2 is on chromosome 3 and SNCA on chromosome 4, and CRISPRi does not cut DNA, excluding a

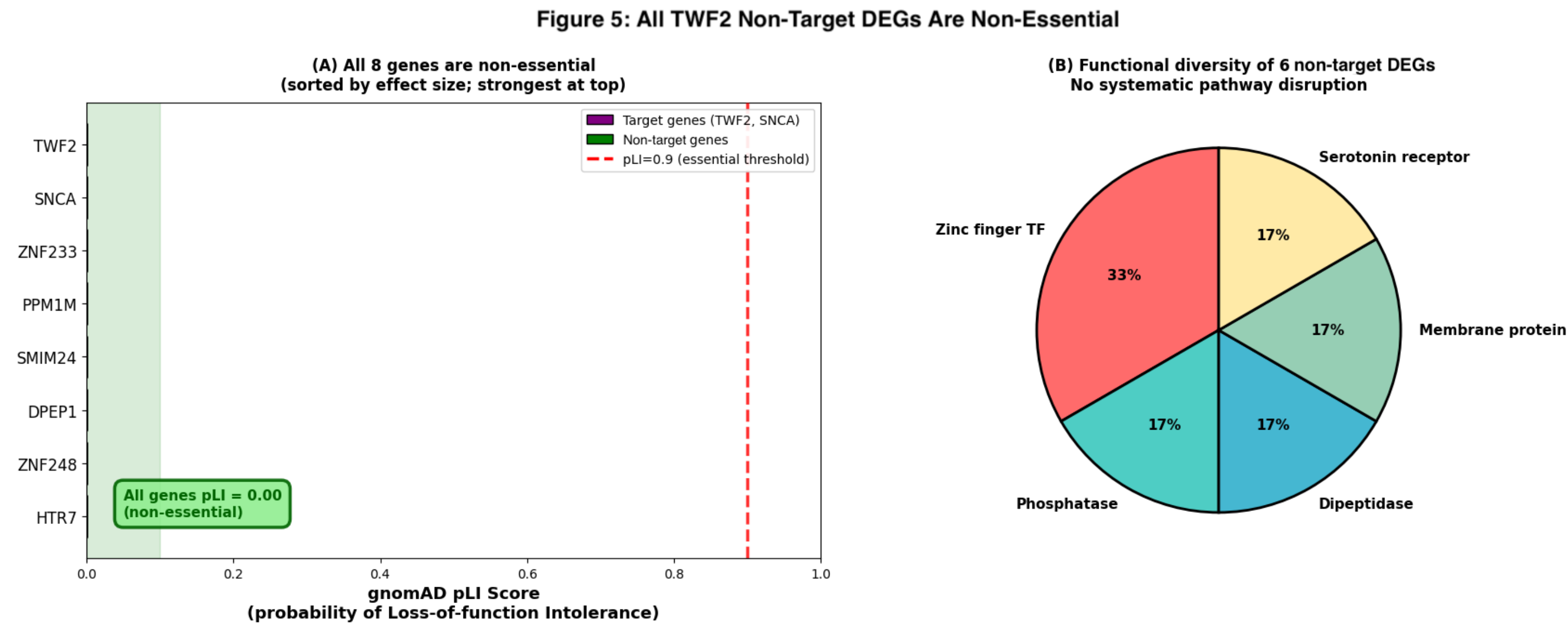


Figure 5. Loss-of-function tolerance of the affected genes. (A) gnomAD pLI scores for the eight TWF2-affected genes; all score 0.00, below the 0.9 threshold for loss-of-function intolerance. (B) The six non-target genes span unrelated functional categories with no shared pathway.

Table 1. Significance and safety of genes affected by TWF2 knockdown

Gene	SNCA Reduction %	log ₂ FC	pLI	Function
SNCA	94.2	-4.11	0.00	α -Synuclein (target)
TWF2	91.1	-3.54	0.00	Actin-binding (target)
ZNF233	76.5	-2.09	0.00	Zinc finger TF
PPM1M	45.2	-0.87	0.00	Phosphatase
SMIM24	37.3	-0.67	0.00	Membrane protein
DPEP1	33.1	-0.58	0.00	Dipeptidase
ZNF248	31.8	-0.55	0.00	Zinc finger TF
HTR7	31.1	-0.54	0.00	Serotonin receptor

5 Discussion

- TWF2 matched direct SNCA knockdown in potency (91.1% vs 94.2%) while altering 4.9x fewer genes.
- TWF2 caps actin filaments and belongs to the cofilin family, which protects against alpha-synuclein toxicity; it also functions in neurite outgrowth and dendritic spine length.
- Only 8 of more than 11,000 genes changed, arguing for a targeted mechanism over broad cytoskeletal disruption.
- TWF2 knockout mice are viable, humans tolerate TWF2 loss-of-function, and a small molecule already targets the TWF2-YAP interaction.
- **Limitations:** computational only; mRNA not protein; stem cells not neurons; 51 of 300 perturbations analyzed.

CONCLUSION

TWF2 knockdown lowered SNCA mRNA by 91.1% while altering only eight genes, all loss-of-function tolerant. TWF2 is a specific indirect route to reduced alpha-synuclein and a candidate for testing in neuronal models.

Next: TWF2 knockdown in iPSC-derived dopaminergic neurons, then measure α -synuclein protein and aggregation in disease-relevant cells.

Code & data: github.com/isobutane101/twf2-snca-notebook
Data: Arc Institute Virtual Cell Challenge.